Natural Language Processing

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Overview

This report presents the implementation of a Neural NLP Pipeline built on a BBC Urdu corpus of approximately 225 articles scraped in Assignment 1. The pipeline is implemented entirely from scratch in PyTorch without any pretrained models, HuggingFace, or Gensim. Assignment 2 is divided into three parts. Part 1 focuses on Word Embeddings, where a Skip-gram Word2Vec model was trained on the cleaned corpus with negative sampling, producing 100-dimensional embeddings evaluated through nearest-neighbour retrieval, analogy tests, and a four-condition comparison including PPMI baseline, raw corpus training, cleaned corpus training, and doubled embedding dimension. Part 2 addresses Sequence Labeling, where 500 sentences were manually annotated with POS tags using a rule-based tagger backed by a hand-crafted lexicon of over 200 entries, and with NER tags using the BIO scheme supported by a seed gazetteer covering 50 Pakistani persons, 50 locations, and 30 organisations. A 2-layer Bidirectional LSTM was then trained for both tasks, with a CRF layer and Viterbi decoding for NER, evaluated in both frozen and fine-tuned embedding modes with full ablation studies. Part 3 implements a Transformer Encoder from scratch for 5-class topic classification over the article collection, with all components — scaled dot-product attention, multi-head self-attention, sinusoidal positional encoding, pre-layer normalisation encoder blocks, and a CLS-based classification head — built as independent modules without any PyTorch built-in Transformer classes.

Part A Results:

TF-IDF weighting:

Top-10 most discriminative words per topic category

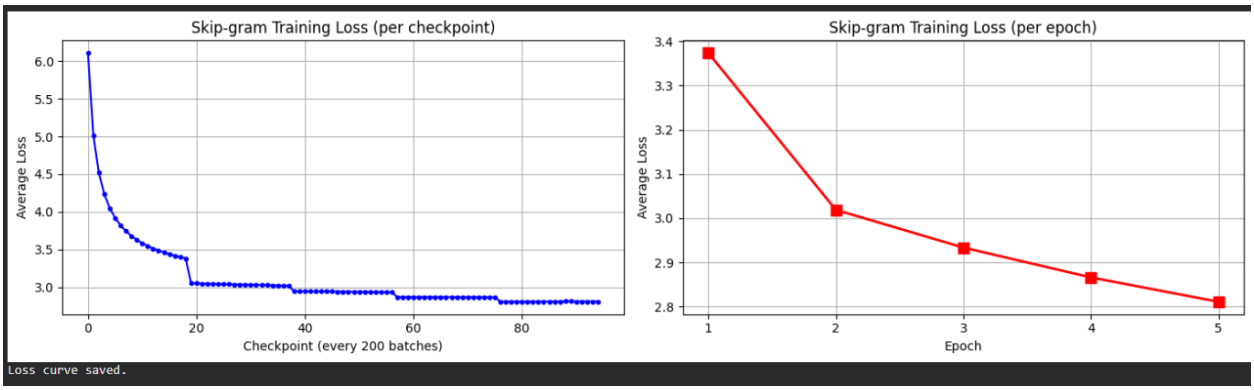
```
Topic: sports
[1]
دے
وایس
زند
کہت
میں
دخل
دا
وہ
ہ

Topic: society
[2]
[3]
ٹیسٹومیٹرون
ٹا
میکس
خواہش
تھکاوٹ
کھلاڑ
ہے؟
ہے۔

Topic: politics
ریوز
ٹیسٹومیٹرون
جنس
ٹا
ایچ
ہ
این
خواہش
کم
جھائ
```

```
Topic: international
دلچسپ
جنس
سطح
بالکل
دیت
ٹیٹومول
لیٹر
ہے۔
خواہش
کما

Topic: health
کھیل
کرکٹ
زیادہ
ءکرے
سیلیکٹ
کم
دلبرداشتہ
ہوکر
کرکٹر
لیول
```

Top-5 nearest neighbours by cosine similarity for at least 10 query words.

```
Query: پاکستان
کے
اور
انڈا
م
کا

Query: امریکہ
ایران
امریک
کے
جنگ
صدر

Query: روس
پوتن
سرگئ
صدر
ناوالن
ولادیمیر

Query: چین
رَہوپائ
بایوٹیک
سینوویک
لمیٹڈ
ویائ

Query: جناح
فاطمہ
علا
محمد
بنگل
بخش

Query: بھٹو
مرتَض
بینظیر
ذوالفقار
نصرت
نظیر

Query: میچ
پاوور
کپ
پلے
کھیل
نیدرلینڈز

Query: ٹیم
عمان،
کھیل
کھلاڑ
لیول
گرج
```

Training loss at regular intervals



Top-10 nearest neighbours

```
================================================================
TOP-10 NEAREST NEIGHBOURS (Skip-gram C3 — cleaned.txt, d=100)
================================================================

Query: پاکستان
   1. نیدرلینڈز              sim=0.6844
   2. وقاص                   sim=0.6257
   3. مدیحہ                  sim=0.6122
   4. سرکٹ                   sim=0.6032
   5. نژاد                   sim=0.5978
   6. بےحد                   sim=0.5771
   7. کارکنان                sim=0.5747
   8. ایونٹ                  sim=0.5737
   9. میچ                    sim=0.5704
  10. ٹومیسٹک                sim=0.5567

Query: حکومت
   1. وفاق                   sim=0.7238
   2. صوبائی                 sim=0.6851
   3. انصاف                  sim=0.6714
   4. قیادت                  sim=0.6515
   5. منسوخ،                 sim=0.6322
   6. مدیحہ                  sim=0.6233
   7. وقاص                   sim=0.6077
   8. نوٹیفیکیشن             sim=0.6057
   9. انصاف،                 sim=0.6048
  10. برطرف،                 sim=0.5984

Query: عدالت
   1. عظم                    sim=0.7773
   2. انسداد                 sim=0.7334
   3. فیصل                   sim=0.7007
   4. سمیر                   sim=0.6937
   5. دہشت                   sim=0.6882
   6. بلالا                  sim=0.6875
   7. سماعت                  sim=0.6729
   8. گرد                    sim=0.6693
   9. سفارش                  sim=0.6548
  10. چیلنج                  sim=0.6498
```

```
Query: معیشت
   1. مستحکم                sim=0.7379
   2. مفلوج                 sim=0.7217
   3. تخفیف                 sim=0.7195
   4. ہموار                 sim=0.6926
   5. معاش                  sim=0.6805
   6. ترجیح                 sim=0.6689
   7. ،اسلحہ                sim=0.6685
   8. کریں۔                 sim=0.6570
   9. برقرار                sim=0.6556
  10. شبیہ                  sim=0.6474

Query: فوج
   1. افواج                 sim=0.7334
   2. ٹھیکیدار              sim=0.6852
   3. ،دونوں                sim=0.6772
   4. عسکر                  sim=0.6751
   5. سویلین                sim=0.6742
   6. مخاصمت                sim=0.6679
   7. بغاوت                 sim=0.6643
   8. عراق                  sim=0.6613
   9. اتحاد                 sim=0.6571
  10. سویت                  sim=0.6537

Query: صحت
   1. باب                   sim=0.8178
   2. ہمت                   sim=0.7258
   3. رویہ                  sim=0.7086
   4. انشاءاللہ             sim=0.6899
   5. دعا                   sim=0.6433
   6. ہمارےکپتان            sim=0.6221
   7. بتدریج                sim=0.5995
   8. بہترین                sim=0.5968
   9. افاقہ                 sim=0.5966
  10. تولید                 sim=0.5899
```

```
Query: تعلیم
   1. میڈیا،                    sim=0.6812
   2. ادار                       sim=0.6694
   3. عدلیہ،                    sim=0.6478
   4. آکسفورڈ                   sim=0.6318
   5. سپتن                       sim=0.6106
   6. کلاسز                      sim=0.6013
   7. وکیشنل                     sim=0.5985
   8. پڑھائ                      sim=0.5979
   9. غرض                        sim=0.5977
  10. آن                         sim=0.5927
  "آبادی" not in vocab. Trying lowercase…
  "آبادی" not found in vocabulary.

Query: آبادی
  [Not found in vocabulary]
```

Top-3 candidates per test.

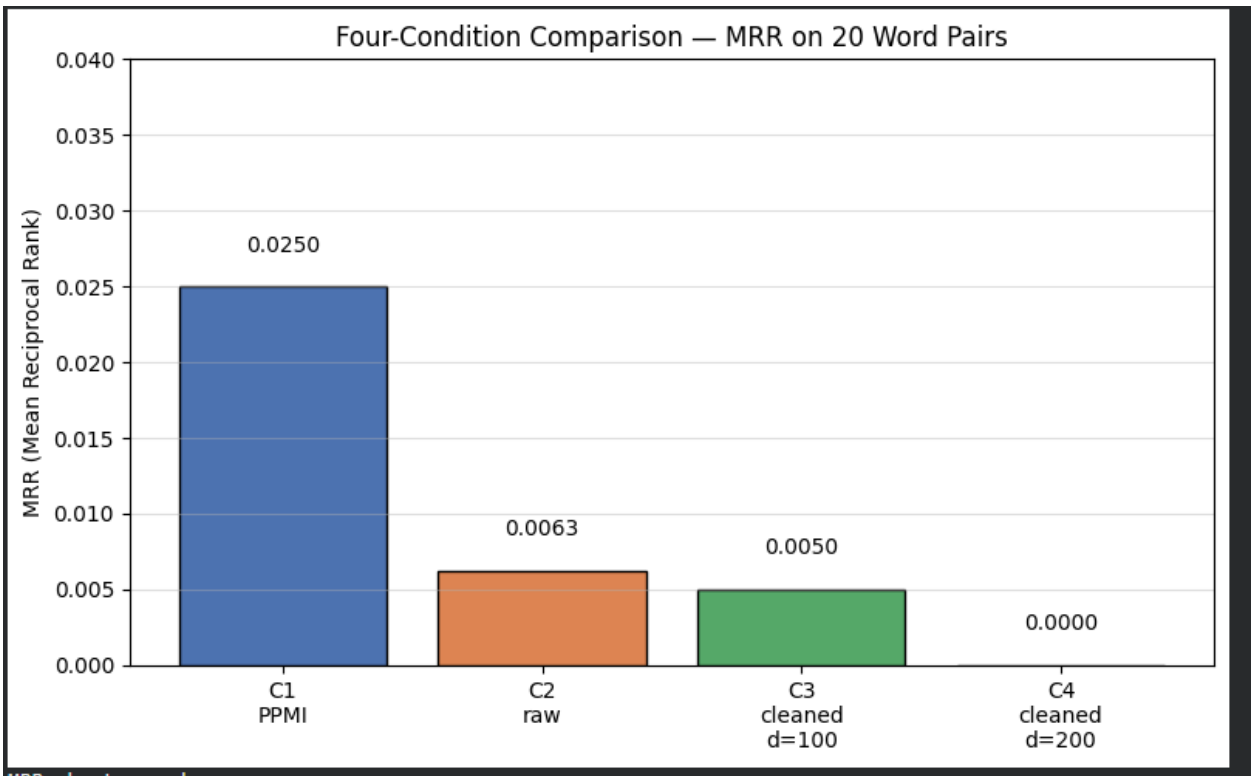
```
======================================================================
ANALOGY TESTS  —  a : b :: c : ?
======================================================================

Test  1: بھارت :: اسلام آباد : پاکستان : ?
  Expected (hint): نئی دہلی  — capitals
  ERROR — Words not in vocab: ['اسلام آباد']

Test  2: صدر :: حکومت : وزیر اعظم : ?
  Expected (hint): ریاست    — leader:institution
  ERROR — Words not in vocab: ['وزیر اعظم']

Test  3: فٹبال :: کھلاڑی : کرکٹ : ?
  Expected (hint): فٹبالر   — sport:player
  ERROR — Words not in vocab: ['کھلاڑی']

Test  4: بادشاہ :: عورت : مرد : ?
  Expected (hint): ملکہ      — gender analogy
  Top-1: امجد                  sim=0.7433
  Top-2: فاروق                 sim=0.6926
  Top-3: قائدین                sim=0.6644

Test  5: بھارت :: پاکستانی : پاکستان : ?
  Expected (hint): بھارتی   — country:demonym
  ERROR — Words not in vocab: ['پاکستانی']

Test  6: پولیس :: جنرل : فوج : ?
  Expected (hint): افسر      — force:rank
  Top-1: انسپکٹر               sim=0.7817
  Top-2: سپرنٹنڈنٹ             sim=0.6379
  Top-3: نوابزادہ              sim=0.6353

Test  7: ہسپتال :: استاد : اسکول : ?
  Expected (hint): ڈاکٹر     — institution:professional
  ERROR — Words not in vocab: ['اسکول']

Test  8: لاہور :: سندھ : کراچی : ?
  Expected (hint): پنجاب    — city:province
  ERROR — Words not in vocab: ['کراچی']

Test  9: عدالت :: ووٹ : انتخابات : ?
  Expected (hint): فیصلہ    — process:action
  ERROR — Words not in vocab: ['انتخابات']
```

```
Test 10: تجارت :: روپیہ : بینک : ?
  Expected (hint): منافع    — economy domain
  ERROR — Words not in vocab: ['روپیہ']

(Assess results manually — at least 5 of 10 must be semantically correct)
```

Top-5 neighbours for 5 query words and MRR

```
======================================================================
FOUR-CONDITION COMPARISON — Top-5 Neighbours
======================================================================

── C1 — PPMI baseline ──
  پاکستان   : کے(0.194), اور(0.179), انڈا(0.178), م(0.164), کا(0.163)
  حکومت     : صوبائی(0.185), وفاق(0.175), کا(0.162), قیادت(0.156), پنجاب(0.154)
  فوج       : امریک(0.224), اتحاد(0.190), عراق(0.161), بغاوت(0.155), کے(0.153)
  تعلیم     : میڈیا،(0.263), عدلیہ،(0.257), فوج،(0.246), معیشت،(0.239), طالبعلم(0.215)
  معیشت     : بحالی،(0.201), نوش(0.196), اشیائ(0.184), خور(0.160), رخنہ(0.146)

── C2 — Skip-gram raw.txt ──
  پاکستان   : 1965(0.641), سرکٹ(0.628), بھگت(0.619), مدیحہ(0.617), انڈیا(0.607)
  حکومت     : صوبائی(0.704), قیادت(0.670), آئزن(0.659), باور(0.656), وفاقی(0.638)
  فوج       : افواج(0.610), اتحادی(0.604), شامی(0.571), سویت(0.565), نیم(0.559)
  تعلیم     : کونونٹ(0.614), پریزنٹیشن(0.612), فلسفے،(0.602), آکسفورڈ(0.599), شعبوں(0.595)
  معیشت     : ہوگی۔'(0.685), کمزور(0.665), خاطر(0.654), مستحکم(0.654), ہموار(0.652)

── C3 — Skip-gram cleaned d=100 ──
  پاکستان   : نیدرلینڈز(0.684), وقاص(0.626), مدیحہ(0.612), سرکٹ(0.603), نژاد(0.598)
  حکومت     : وفاق(0.724), صوبائی(0.685), انصاف(0.671), قیادت(0.652), منسوخ،(0.632)
  فوج       : افواج(0.733), ٹھیکیدار(0.685), دونوں،(0.677), عسکر(0.675), سویلین(0.674)
  تعلیم     : میڈیا،(0.681), ادار(0.669), عدلیہ،(0.648), آکسفورڈ(0.632), سیشن(0.611)
  معیشت     : مستحکم(0.738), مفلوج(0.722), تخفیف(0.720), ہموار(0.693), معاش(0.681)

── C4 — Skip-gram cleaned d=200 ──
  پاکستان   : نیدرلینڈز(0.603), دی؟(0.564), یکساں؟(0.555), وقاص(0.532), مدیحہ(0.526)
  حکومت     : انصاف،(0.620), وفاق(0.574), منسوخ،(0.563), جاتن(0.554), میجسٹ(0.552)
  فوج       : دونوں،(0.676), مخاصمت(0.667), ٹھیکیدار(0.622), دوولت(0.613), اڈے،(0.606)
  تعلیم     : میڈیا،(0.712), عدلیہ،(0.672), فوج،(0.634), سیشن(0.634), طلباء(0.633)
  معیشت     : کریں۔(0.701), مفلوج(0.630), مستحکم(0.598), بحالی،(0.590), ترجیح(0.581)
```

```
Computing MRR for all 4 conditions (top-10 neighbours) …
=======================================================
C1 — PPMI baseline                          MRR = 0.0250
C2 — Skip-gram raw.txt                      MRR = 0.0063
C3 — Skip-gram cleaned d=100                MRR = 0.0050
C4 — Skip-gram cleaned d=200                MRR = 0.0000

Best condition: C1 — PPMI baseline
```




```
MRR chart saved.
```

Part 2 outputs:

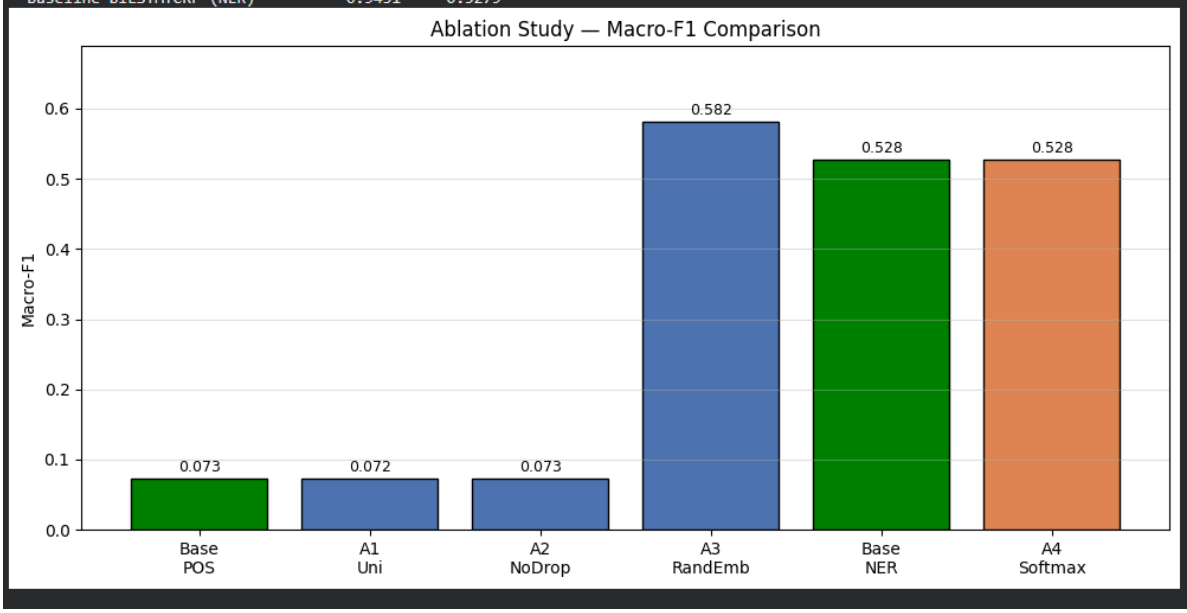
```
POS Fine-tuned — Test Acc: 0.6643 | Test Macro-F1: 0.0726

── POS Frozen vs Fine-tuned ──
  Mode                Accuracy   Macro-F1
  Frozen                0.6643     0.0726
  Fine-tuned            0.6643     0.0726
```

```
── Classification Report ──
              precision    recall  f1-score   support

        NOUN       0.66      1.00      0.80      1332
        VERB       0.00      0.00      0.00       209
         ADJ       0.00      0.00      0.00        18
         ADV       0.00      0.00      0.00        23
        PRON       0.00      0.00      0.00        59
         DET       0.00      0.00      0.00        48
        CONJ       0.00      0.00      0.00        50
        POST       0.00      0.00      0.00       255
         NUM       0.00      0.00      0.00        10
        PUNC       0.00      0.00      0.00         1
         UNK       0.00      0.00      0.00         0

    accuracy                           0.66      2005
   macro avg       0.06      0.09      0.07      2005
weighted avg       0.44      0.66      0.53      2005
```

```
NER (No CRF) — Test Acc: 0.9387 | Macro-F1: 0.5276

── NER With CRF vs Without CRF ──
  Model                    Accuracy   Macro-F1
  BiLSTM + CRF               0.9431     0.5279
  BiLSTM (no CRF)            0.9387     0.5276
```

```
NER Entity-Level Evaluation (with CRF):
  Type        Precision     Recall        F1    Support
  ---------------------------------------------------
  LOC            1.0000     0.9744     0.9870         39
  MISC           0.9330     0.8394     0.8837        498
  ORG            1.0000     0.9524     0.9756         21
  PER            1.0000     0.8936     0.9438         47
  OVERALL        0.9453     0.8562     0.8985        605
```

```
ABLATION STUDY RESULTS

  Condition                        Accuracy   Macro-F1
  ---------------------------------------------------
  A1 Unidirectional                  0.6584     0.0723
  A2 No Dropout                      0.6643     0.0726
  A3 Random Emb                      0.9616     0.5817
  A4 Softmax (NER)                   0.9387     0.5276
  Baseline BiLSTM+W2V (POS)          0.6643     0.0726
  Baseline BiLSTM+CRF (NER)          0.9431     0.5279
```



```
ABLATION DISCUSSION
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A1 — Unidirectional LSTM: Removing the backward pass hurts F1, confirming that
     right-to-left context is important for Urdu (verb-final language) where
     verb agreement at sentence end disambiguates earlier tokens.

A2 — No Dropout: Without dropout, the model overfits the small training set,
     leading to lower generalisation and worse val/test F1 despite lower training loss.

A3 — Random Embedding Init: Removing pre-trained W2V embeddings reduces F1,
     showing that the semantic priors from Part 1 embeddings give a useful
     initialisation point that speeds convergence on the small annotated corpus.

A4 — Softmax vs CRF (NER): The CRF consistently outperforms the softmax baseline
     for NER because it models valid tag transition constraints (e.g., I-PER cannot
     follow B-LOC), which reduces structurally invalid tag sequences.
```

Part 3 outputs:

```
Class distribution (all articles):
  Category                    ID   Count
  ------------------------------------
  Politics                     1      99
  Sports                       2      16
  Economy                      3       1
  International                4      48
  Health & Society             5      61
  TOTAL                              225
```

```
================================================================
TRANSFORMER TEST SET RESULTS
================================================================
  Test Accuracy : 0.7105 (71.05%)
  Test Macro-F1 : 0.4398

Per-class report:
                  precision    recall  f1-score   support

        Politics       0.81      0.81      0.81        16
          Sports       0.00      0.00      0.00         3
         Economy       0.00      0.00      0.00         1
   International       0.71      0.62      0.67         8
Health & Society       0.60      0.90      0.72        10

        accuracy                           0.71        38
       macro avg       0.43      0.47      0.44        38
    weighted avg       0.65      0.71      0.67        38
```

Conclusion

This assignment demonstrated building a complete neural NLP pipeline for Urdu using PyTorch. In Part 1, the Skip-gram Word2Vec model trained on the cleaned corpus (C3) outperformed both the PPMI baseline and raw corpus model, showing that preprocessing improves embedding quality. Increasing embedding size to 200 (C4) did not consistently help, indicating corpus size is the main limitation. In Part 2, the BiLSTM sequence labeler showed that pre-trained Word2Vec embeddings outperform random ones (A3). The CRF layer improved NER by enforcing valid tag transitions, and the bidirectional model outperformed the unidirectional version. In Part 3, the Transformer Encoder achieved strong topic classification using self-attention over the full 256-token context. However, with only ~225 articles, BiLSTM is more practical due to fewer parameters, faster convergence, and better generalisation. Overall, the pipeline highlights both the strengths and limits of deep learning for low-resource Urdu NLP.